WORKED SOLUTION

STEP II 2003, Question 13

Probability and Statistics • Truncated Poisson Distribution

STEP II 2003, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q13

The key idea

This is a Poisson distribution with the value 0 removed and the remaining probabilities scaled up so they still sum to 1. Every sum we need is therefore a Poisson sum with its $k = 0$ term missing, and for the mean and variance the $k = 0$ term contributes nothing anyway. So we can borrow the familiar Poisson facts $\sum k\,p_k = \lambda$ and $\sum k^2 p_k = \lambda + \lambda^2$ and simply multiply by A.

Solution

The constant A

Write $p_k = \dfrac{\lambda^k \mathrm{e}^{-\lambda}}{k!}$ for the ordinary $\mathrm{Po}(\lambda)$ probabilities. Since $\sum_{k\geq 0} \lambda^k/k! = \mathrm{e}^{\lambda}$,

$$\sum_{k=1}^{\infty} p_k = \mathrm{e}^{-\lambda}(\mathrm{e}^{\lambda} - 1) = 1 - \mathrm{e}^{-\lambda}.$$

The probabilities $\mathrm{P}(X = k) = Ap_k$ must sum to 1, so $A(1 - \mathrm{e}^{-\lambda}) = 1$, that is,

$$A = (1 - \mathrm{e}^{-\lambda})^{-1},$$

as required. Note that $0 < 1 - \mathrm{e}^{-\lambda} < 1$ for $\lambda > 0$, so $A > 1$: the leftover probability p_0 has been shared out among $k \geq 1$.

The mean

The $k = 0$ term of $\sum k\,p_k$ is zero, so

$$\sum_{k=1}^{\infty} k\,p_k = \sum_{k=0}^{\infty} k\,p_k = \lambda,$$

the mean of $\mathrm{Po}(\lambda)$. (Directly: $\sum_{k\geq 1} k\lambda^k \mathrm{e}^{-\lambda}/k! = \lambda \sum_{k\geq 1} \lambda^{k-1}\mathrm{e}^{-\lambda}/(k-1)! = \lambda$.) Hence

$$\mu = \mathrm{E}(X) = A\lambda = \frac{\lambda}{1-\mathrm{e}^{-\lambda}}.$$

The variance

In the same way, the $k = 0$ term of $\sum k^2 p_k$ is zero, and for $\mathrm{Po}(\lambda)$ we have $\mathrm{E}(Y^2) = \mathrm{Var}(Y) + (\mathrm{E}\,Y)^2 = \lambda + \lambda^2$. If you'd rather not quote that, write $k^2 = k(k-1) + k$:

$$\sum_{k\geq 2} k(k-1)\frac{\lambda^k \mathrm{e}^{-\lambda}}{k!} = \lambda^2 \sum_{k\geq 2} \frac{\lambda^{k-2}\mathrm{e}^{-\lambda}}{(k-2)!} = \lambda^2, \qquad \text{so} \qquad \sum_{k\geq 1} k^2 p_k = \lambda^2 + \lambda.$$

Therefore

$$\begin{aligned} \mathrm{E}(X^2) &= A\lambda(1+\lambda) = \mu(1+\lambda), \\ \mathrm{Var}(X) &= \mathrm{E}(X^2) - \mu^2 = \mu(1+\lambda) - \mu^2 = \mu(1-\mu+\lambda), \end{aligned}$$

as required.

The inequalities

Ask yourself Where could a strict inequality $\mu < 1 + \lambda$ come from?

From the variance. X takes every value $1, 2, 3, \ldots$ with positive probability, so it is not constant, and a non-constant random variable has strictly positive variance (as $\mathrm{Var}(X) = \mathrm{E}((X-\mu)^2)$ and $(X-\mu)^2 > 0$ with positive probability). So

$$\mu(1-\mu+\lambda) > 0.$$

Since $\mu > 0$ (it is the mean of a positive random variable), we may divide by μ to get $1 - \mu + \lambda > 0$, i.e. $\mu < 1 + \lambda$.

For the lower bound, $\mu = A\lambda$ with $A > 1$ and $\lambda > 0$, so $\mu > \lambda$. Together,

$$\lambda < \mu < 1 + \lambda.$$

Normal approximation with $\lambda = 100$

Now e^{-100} is astronomically small (about 4×10^{-44}), so to any sensible accuracy

$$\mu = \frac{100}{1-\mathrm{e}^{-100}} \approx 100, \qquad \mathrm{Var}(X) = \mu(101-\mu) \approx 100 \times 1 = 100.$$

So we approximate X by $Y \sim \mathrm{N}(100, 10^2)$. Since X is integer-valued, we use a continuity correction: the event $X = 100$ corresponds to $99.5 < Y < 100.5$. Standardising with $Z = (Y - 100)/10$,

$$\begin{aligned} \mathrm{P}(X = 100) &\approx \mathrm{P}(-0.05 < Z < 0.05) \\ &= 2\Phi(0.05) - 1 \\ &\approx 2(0.51994) - 1 = 0.0399, \end{aligned}$$

so $\mathrm{P}(X = 100) \approx 0.04$ to 2 decimal places.

Checks

- As $\lambda \to 0$, $\mu = \lambda/(1 - \mathrm{e}^{-\lambda}) \to 1$ and the variance $\to 0$: almost all the probability sits at $k = 1$, as it should. This matches the inequality $\lambda < \mu < 1 + \lambda$ squeezing μ towards 1.
- Summing the series numerically for many random λ reproduces A, μ and $\mu(1 - \mu + \lambda)$ to machine accuracy.
- For $\lambda = 100$ the exact probability is $\dfrac{100^{100}\mathrm{e}^{-100}}{100!\,(1 - \mathrm{e}^{-100})} \approx 0.03986$, against the normal approximation 0.03988. Both round to 0.04.

Common slips

- Forgetting the continuity correction and computing $\mathrm{P}(Y = 100) = 0$, or using a density value without justification.
- Proving only $\mu \leq 1 + \lambda$: the strict inequality needs the observation that X is not constant.
- Dividing by μ without noting that $\mu > 0$.

Beyond STEP

X is a *zero-truncated Poisson* distribution, which statisticians use when a count of zero can't be observed (for example, the number of occupants of cars passing a checkpoint: no car has zero occupants). The normal approximation works because a $\mathrm{Po}(100)$ variable is the sum of 100 independent $\mathrm{Po}(1)$ variables, so the central limit theorem applies; the truncation is invisible at this scale. Grimmett and Welsh, *Probability: An Introduction*, covers generating functions and the central limit theorem behind both halves of the question.

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